HAMID KHABIRI, PhD

ROBOTICS ENGINEER | MECHANICAL DESIGN · MOTION CONTROL · SYSTEM INTEGRATION

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PROFESSIONAL SUMMARY

Robotics engineer with 8+ years of combined academic and industry experience developing electromechanical systems from concept to functional prototype. Hands-on expertise in mechanical design, manufacturing, electrical integration, real-time motion control and safety verification.

CORE TECHNICAL SKILLS

Mechanical design & manufacturing: SolidWorks assemblies and drawings; mechanism design; engineering calculations; tolerances; design for manufacturing (DFM); materials and process selection; CNC machining and 3D-printed prototypes.

Motion control & software: TwinCAT 3 (Structured Text, PLC/NC, HMI TE2000); EtherCAT/PDO; position, velocity and torque control; Unity game design and robot integration; C++/MFC and MATLAB.

Electromechanical integration & safety: Mechanical/electrical fabrication and assembly; servo systems, encoders and sensors; electrical schematics, wiring and power distribution; digital/analog I/O; hazard assessment; hardware/software safety and E-stop/STO interfaces.

ENGINEERING EXPERIENCE

Postdoctoral Fellow - Robotics R&D | University of Alberta • Nov 2024–Present

- Independently conceived, designed and built a custom 3-DOF robotic system, including mechanical architecture, CAD, engineering calculations, tolerances, material selection and manufacturable components.
- Selected actuators, drives, encoders and transmissions; personally fabricated and assembled the mechanical prototype and designed, wired and integrated electrical hardware.
- Programmed TwinCAT 3/EtherCAT position, velocity and torque control, PLC/NC logic, field I/O and HMI; designed and integrated Unity rehabilitation games.
- Conducted hazard assessments, implemented layered hardware/software safety (E-stop/STO), and performed bench verification and preliminary tests with five participants.

Mechanical Designer | Noavaran Sanat Rabin (NSR) • Apr 2023–Sep 2024

- Designed, fabricated and assembled a cell-culture incubator with image monitoring; redesigned, fabricated and assembled elements of a stereotactic head frame.

Mechanical Engineer | Sina Robotics and Medical Innovators • Jan 2019–Dec 2020

- Designed and built surgical-simulator mechanisms for SinaSim; completed engineering calculations, manufacturing drawings, assembly, testing, troubleshooting and design improvements.

SELECTED ROBOTICS PROJECTS

Wearable robotic hand orthosis (PhD): Designed and fabricated an underactuated robotic device, developed a C++/MFC control interface and performed clinical evaluation.

MRI-compatible wrist robot: Designed, fabricated and assembled a 2-DOF pneumatic wrist device for fMRI research.

EDUCATION

PhD, Medical Robotics | Tehran University of Medical Sciences, Tehran, Iran — 2022

MSc, Mechanical Engineering (Biomechanics) | Sahand University of Technology, Tabriz, Iran — 2014

BSc, Biomedical Engineering (Biomechanics) | Islamic Azad University, Science and Research Branch, Tehran, Iran — 2011

PATENTS & RESEARCH

Inventor on published U.S. patent application US20260109022A1 and three Iranian patents; co-author of eight peer-reviewed journal articles.